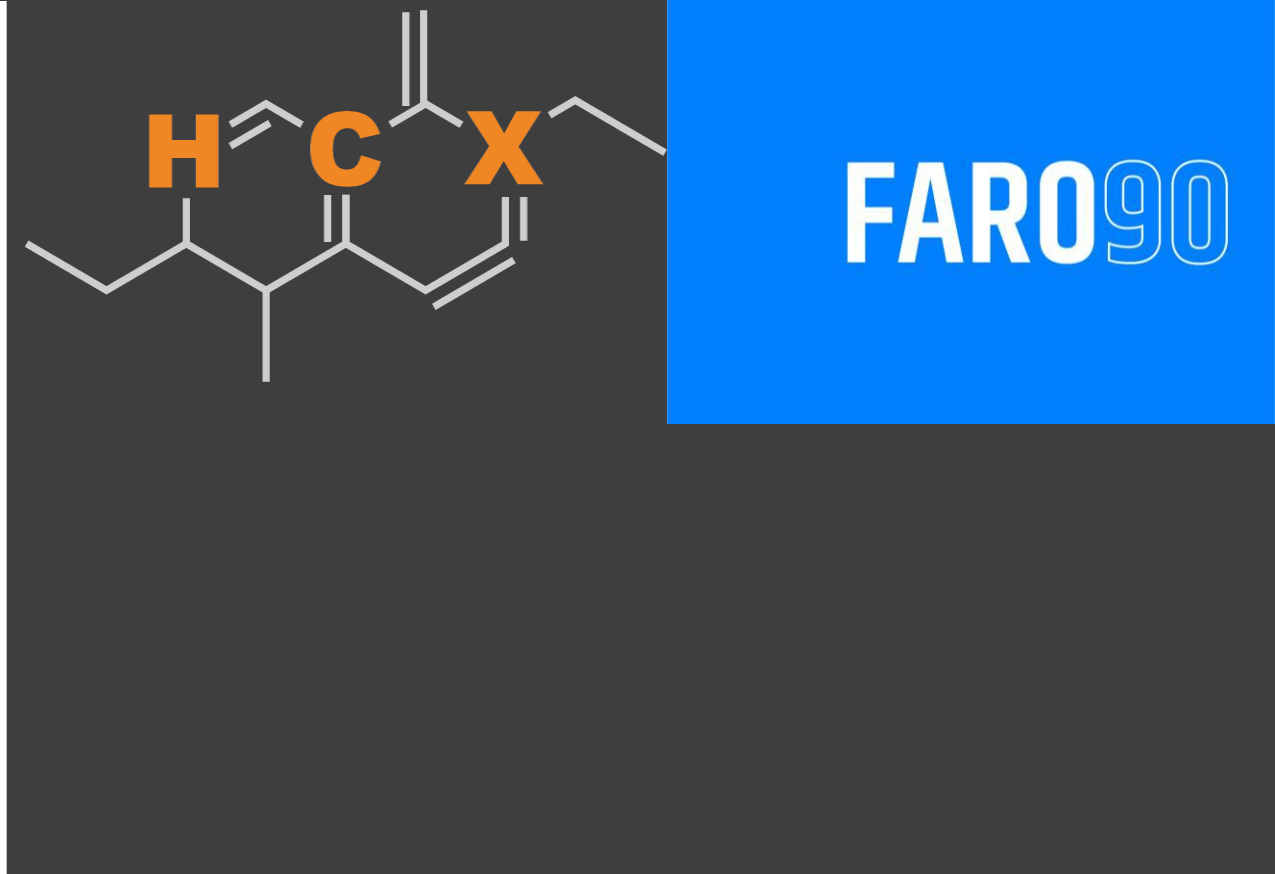




Ethanol Blending in Gasoline - Portugal

September, 2025

Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Portugal

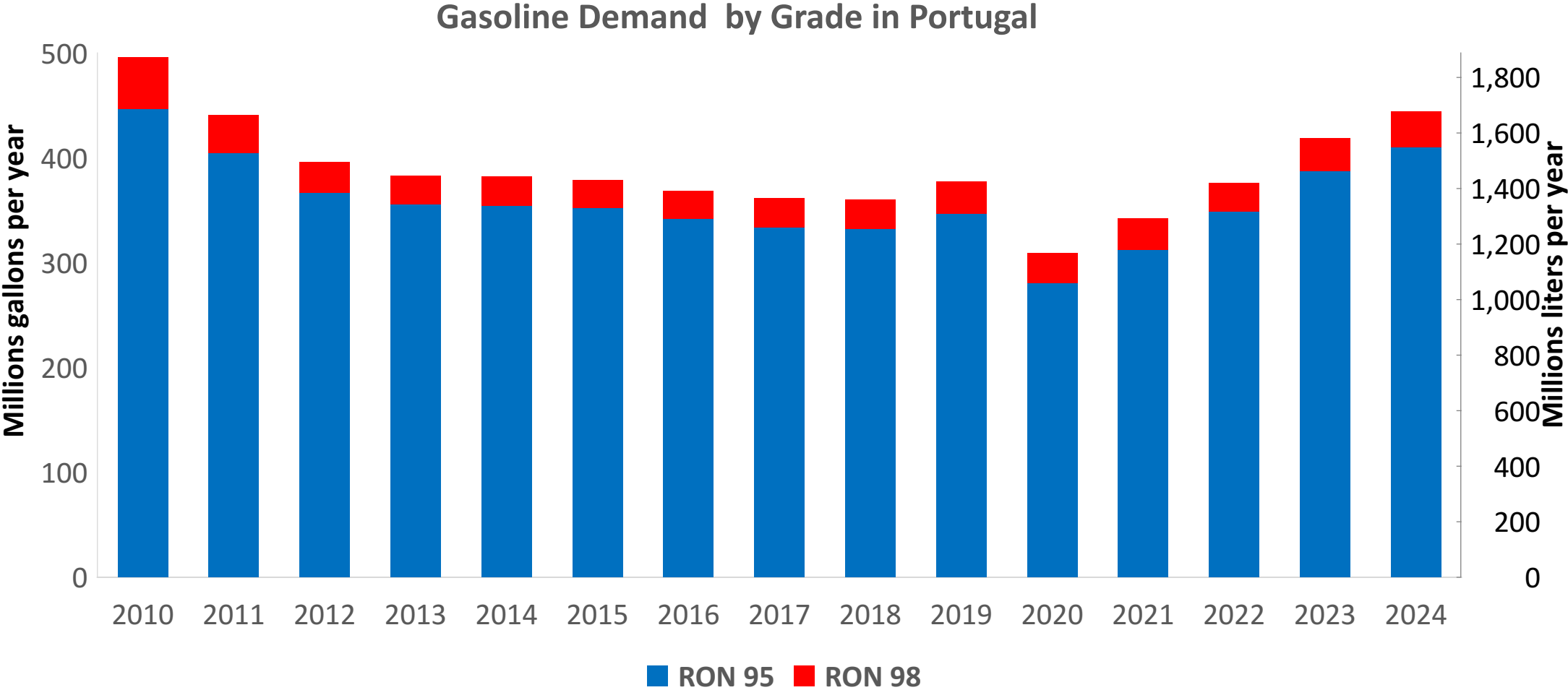


In 2024, gasoline consumption in Portugal was 1,688 million liters (446 million gallons) and growth rate was 0.2%. Gasoline production and net imports were 2,990 and -1,310 million liters (790 and -346 million gallons) correspondingly. Portugal complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Portugal was 64 million liters (17 million gallons) representing 3.8% of gasoline demand. Portugal does not produce ethanol, supply is done through imports. Local production plants are being built.

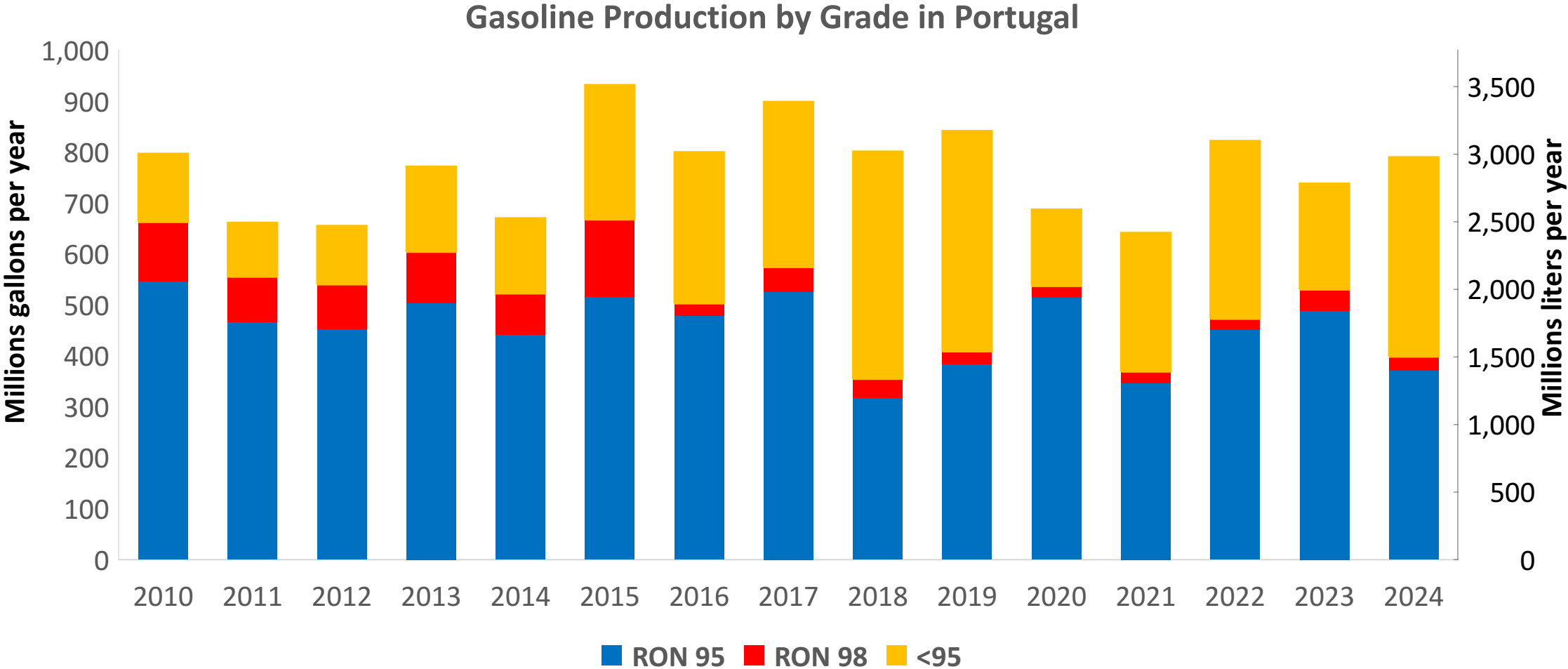
Source: Eurostat, European Commission

Gasoline Demand in Portugal



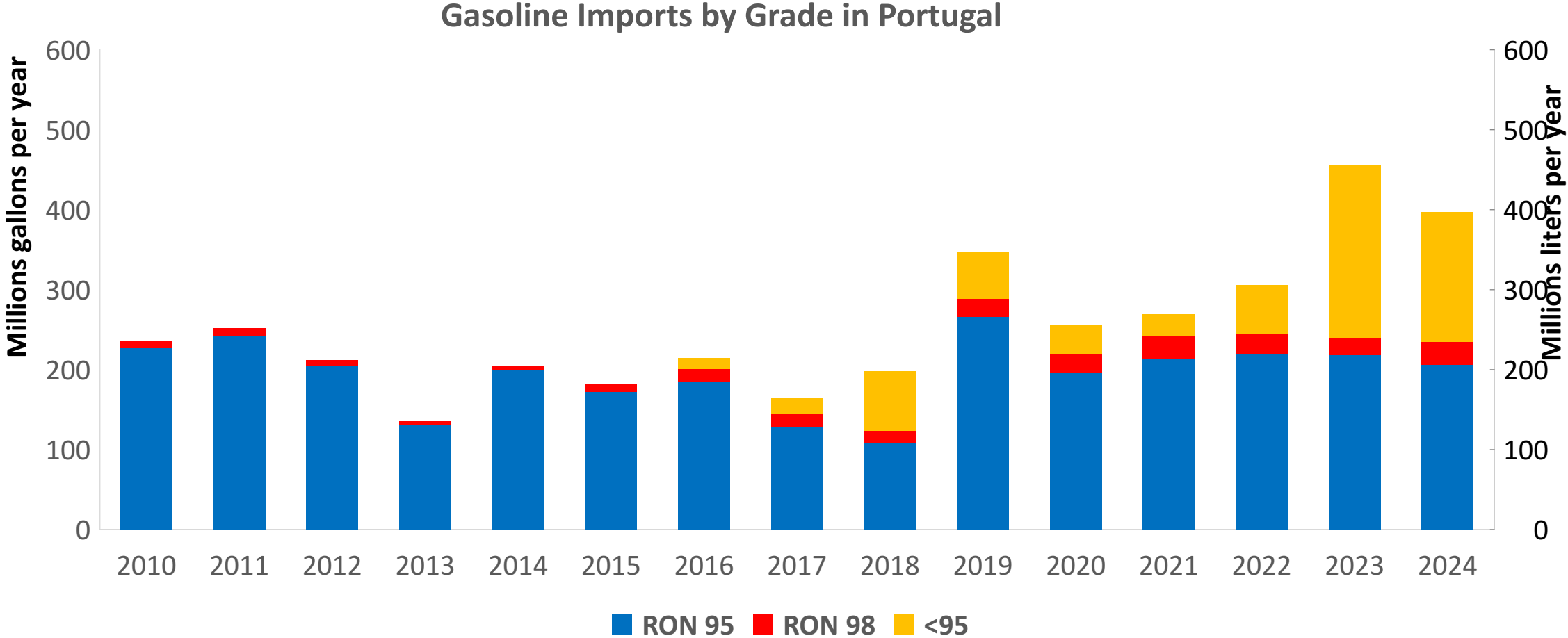
Source: Eurostat

Gasoline Production in Portugal



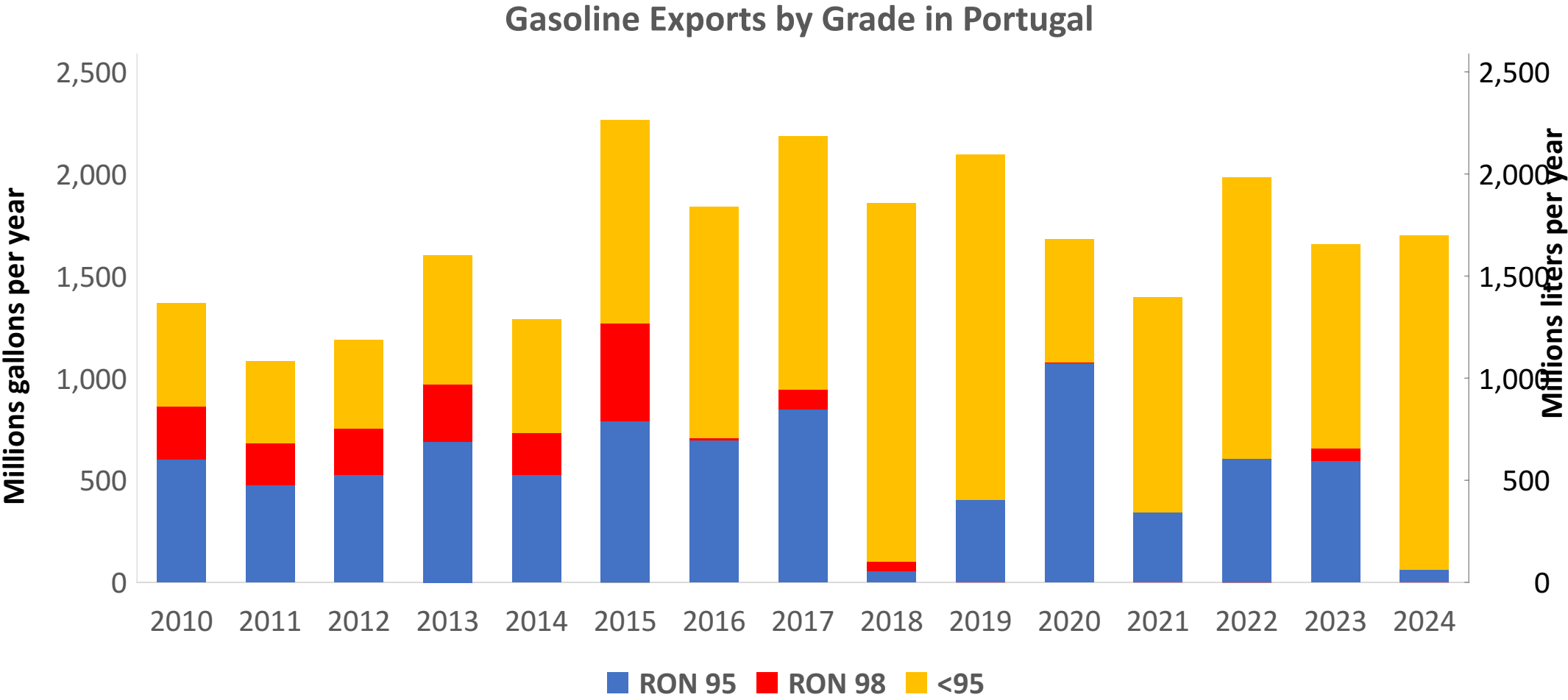
Source: Eurostat

Gasoline Imports in Portugal



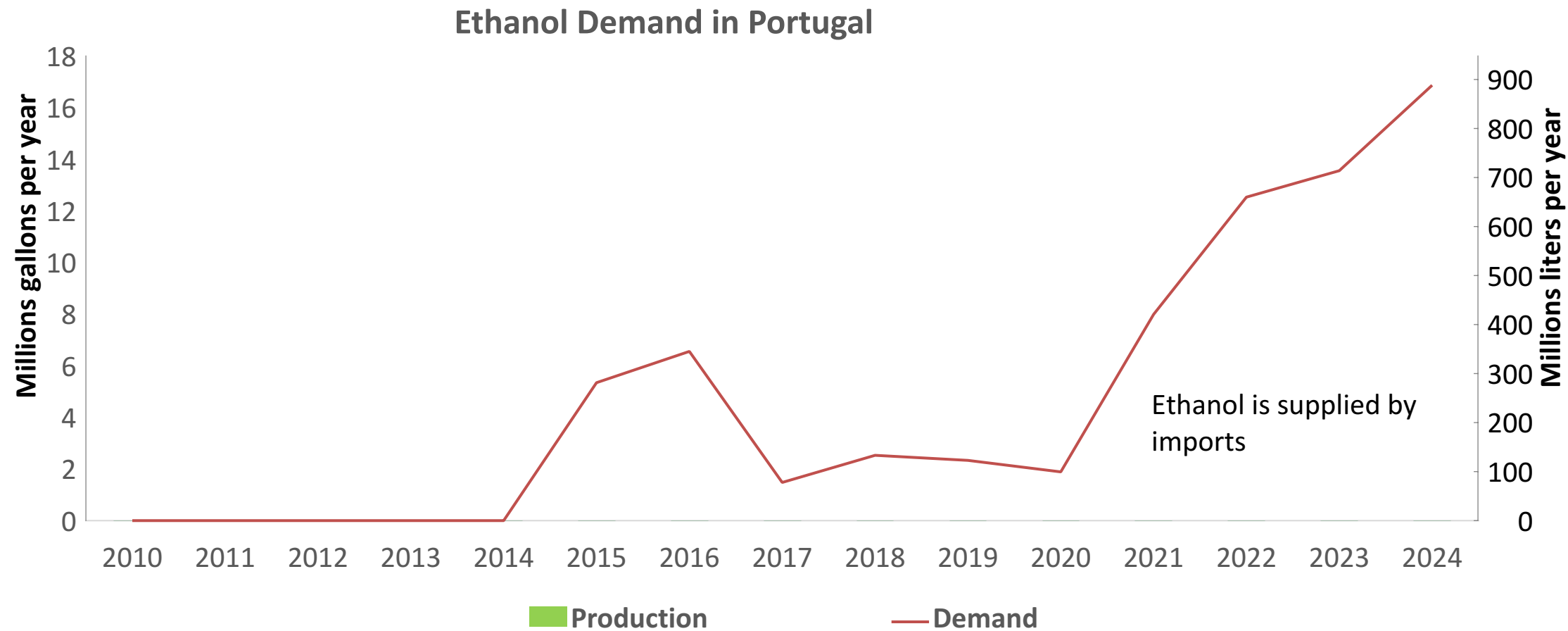
Source: Eurostat

Gasoline Exports in Portugal



Source: Eurostat

Ethanol Balance in Portugal



Gasoline Quality in Portugal

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	13.0							
Advanced Biofuels (%cal)	2.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

The FARO90 logo is shown in a blue box, and a chemical structure is displayed to its right.

Catalytic Gasoline
Light Primary Naphta
Heavy Primary Naphta
Reformate
Isopentane
Alkylate
Isomerate
Hydrocracked Gasoline
Coker Naphta
n-Butane
MTBE
ETBE
TAME
Aromatics

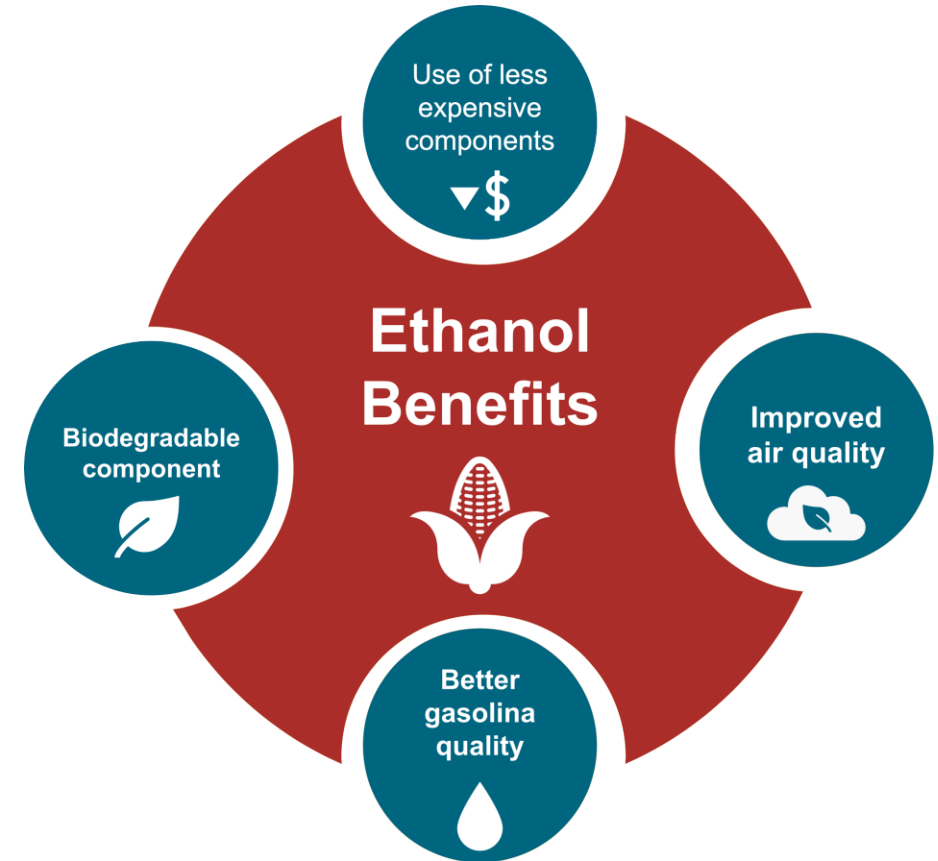
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

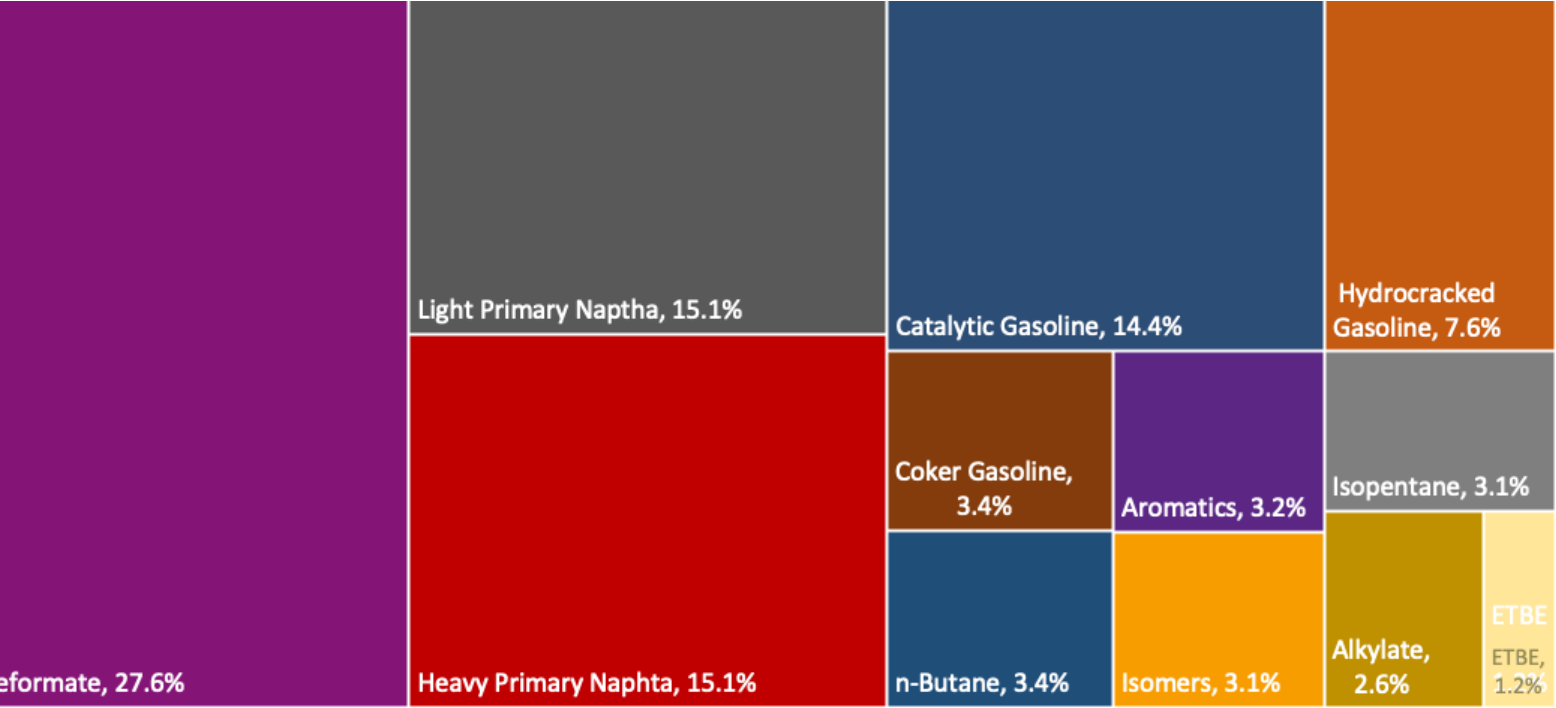
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

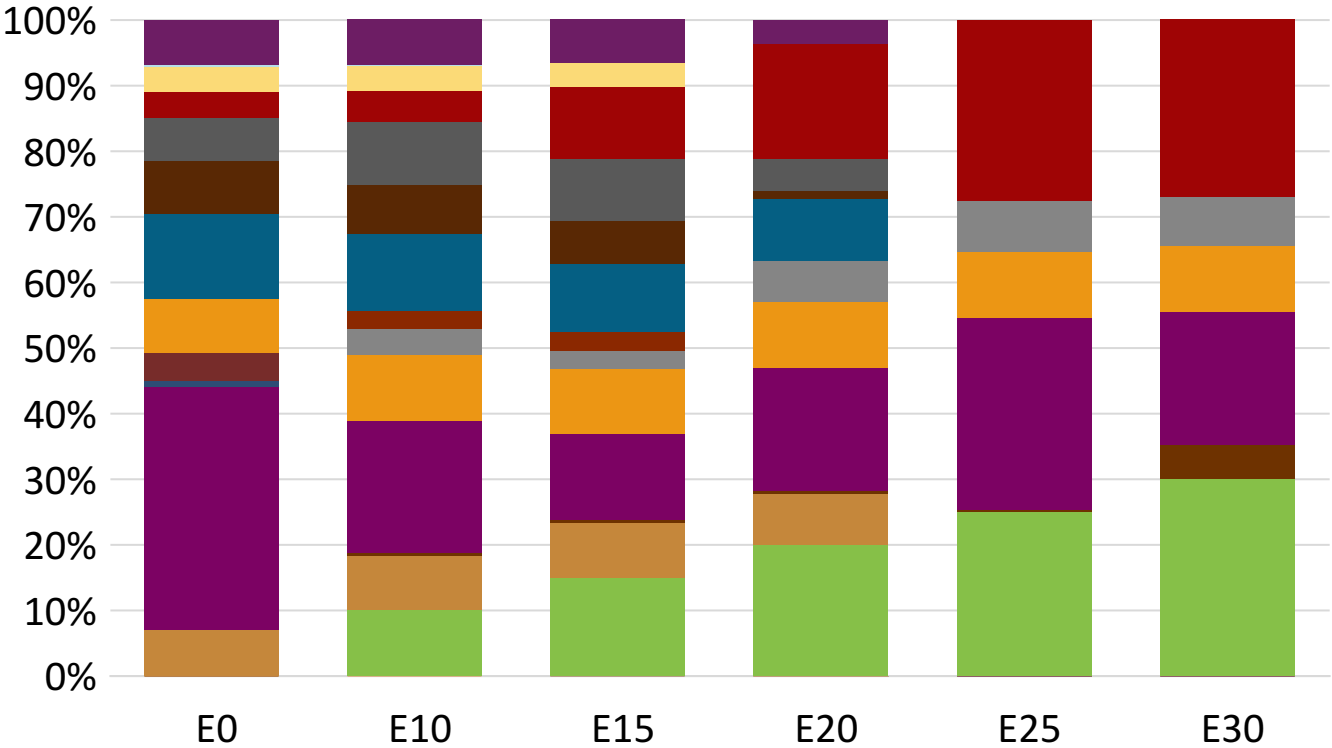
The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Portugal Available Blending Components



Portugal – Gasoline 95 – Constant Octane



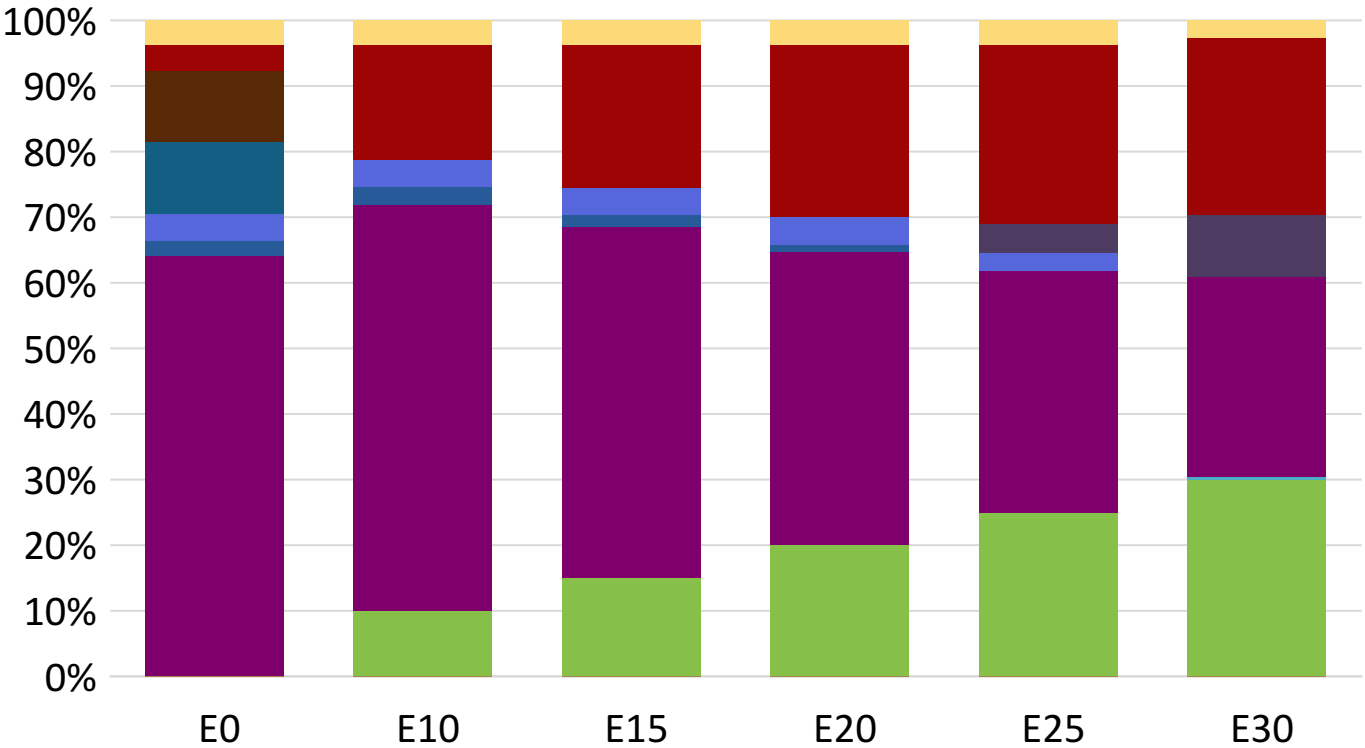
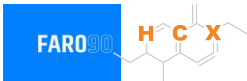
- Ethanol
- Alkylate
- Raffinate
- Reformate
- n-Butane
- Isomerate
- Isopentane
- Hydrocracked Gasoline
- Catalytic Gasoline
- Coker Naphtha
- Light Primary Naphtha
- Heavy Primary Naphtha
- MTBE
- ETBE
- TAME
- Aromatics

Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.3	95.8
Price (US\$/gal)	\$ 2.290	\$ 2.178	\$ 2.098	\$ 2.008	\$ 1.893	\$ 1.830

Source: Faro90

Portugal - Gasoline 98 - Constant Octane

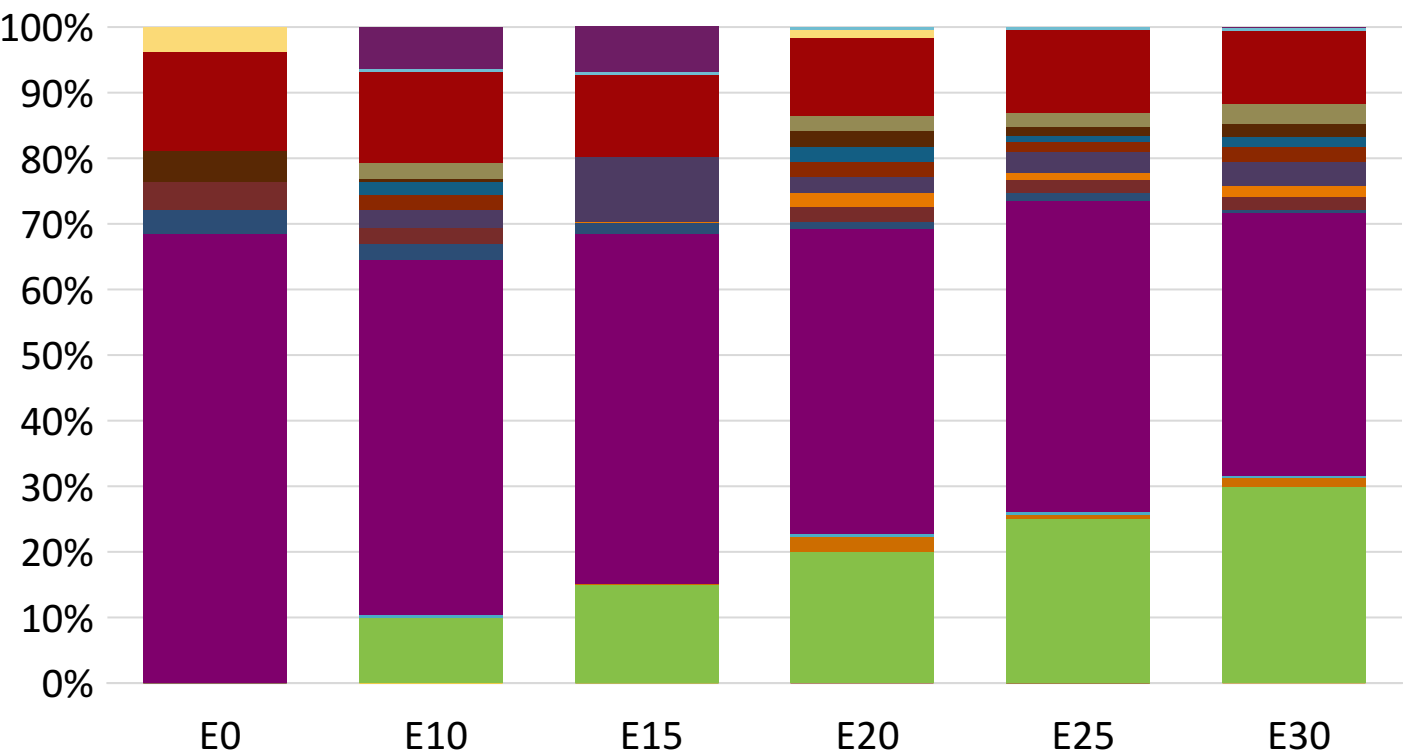


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.359	\$ 2.235	\$ 2.166	\$ 2.091	\$ 2.013	\$ 1.934

Source: Faro90

Portugal – Gasoline 95 – Increased Octane

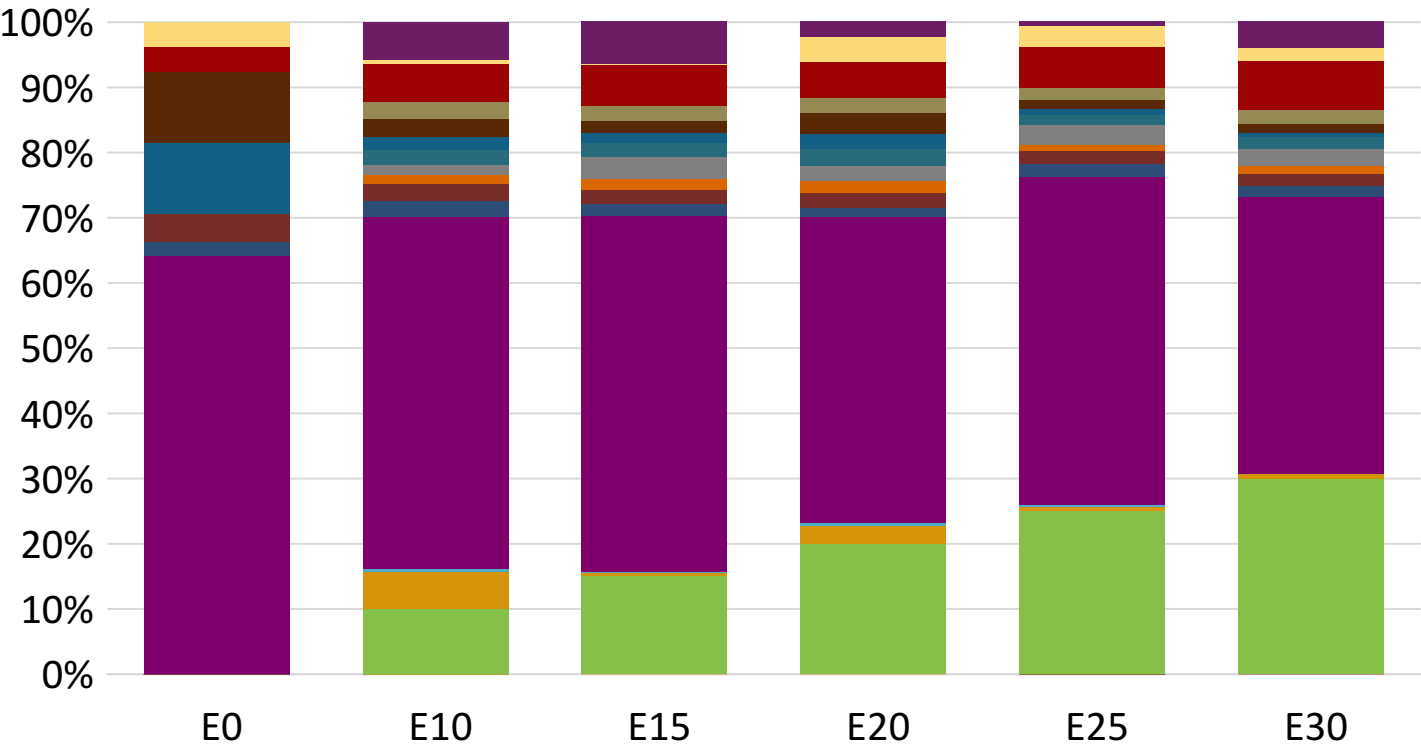


Average CIF 2024 Prices

Octane (RON)	95.0	96.8	98.2	99.9	101.4	102.6
Price (US\$/gal)	\$ 2.222	\$ 2.222	\$ 2.222	\$ 2.222	\$ 2.222	\$ 2.222

Source: Faro90

Portugal – Gasoline 98 – Increased Octane



Average CIF 2024 Prices

Octane (RON)	98.0	99.3	100.9	102.5	104.0	105.0
Price (US\$/gal)	\$ 2.359	\$ 2.359	\$ 2.359	\$ 2.359	\$ 2.359	\$ 2.359

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

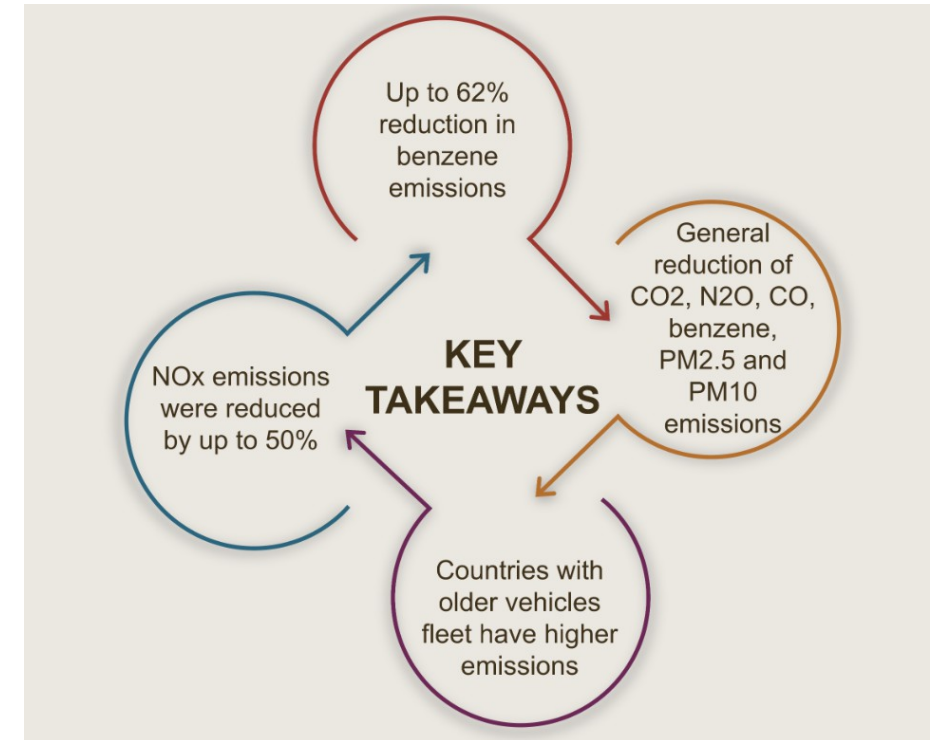
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

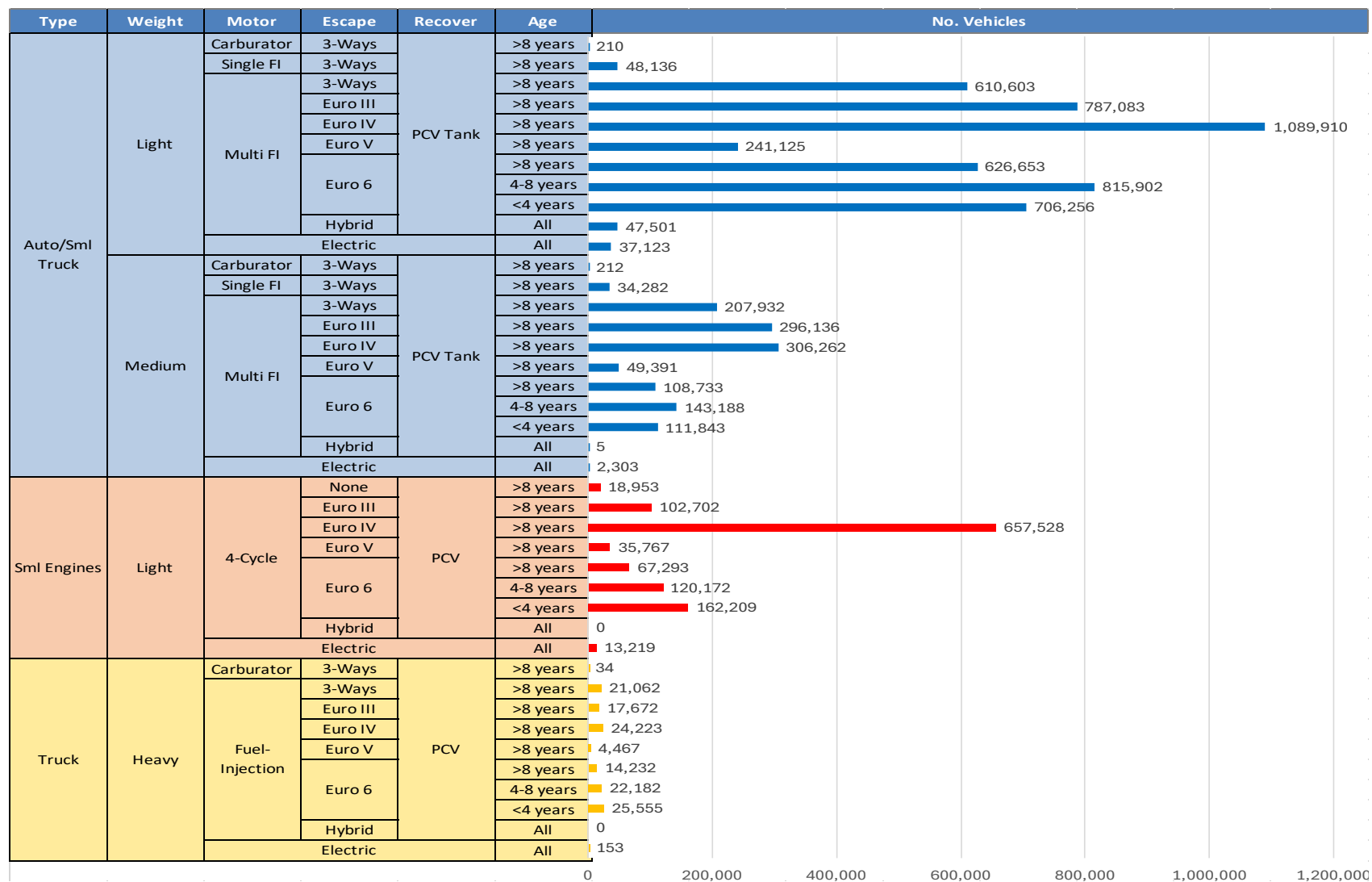
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



*Source: Bureau of transportation statistics.

Gasoline Vehicle Fleet - Portugal

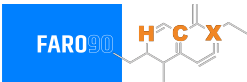


Vehicle Fleet: 7,578,214
Gasoline Vehicle Fleet: 2,654,720

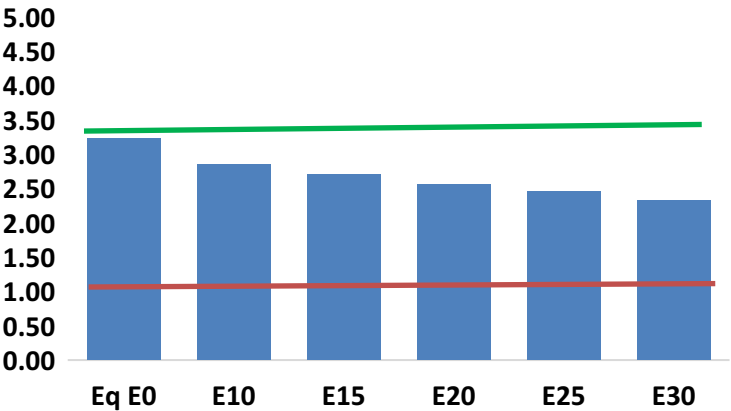
Source: Eurostat, ACEA

Portugal – Vehicular Emissions

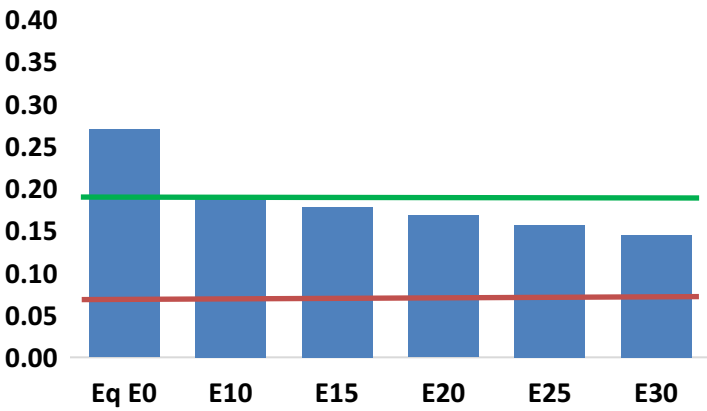
TIER III BIN 125 United States
Euro 6 Standard



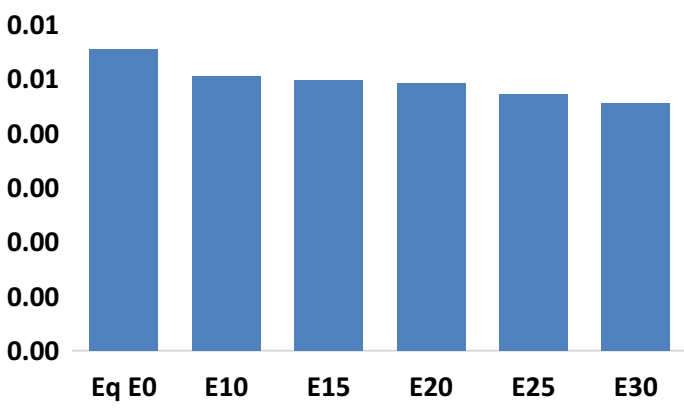
Carbon Monoxide (CO) g/km



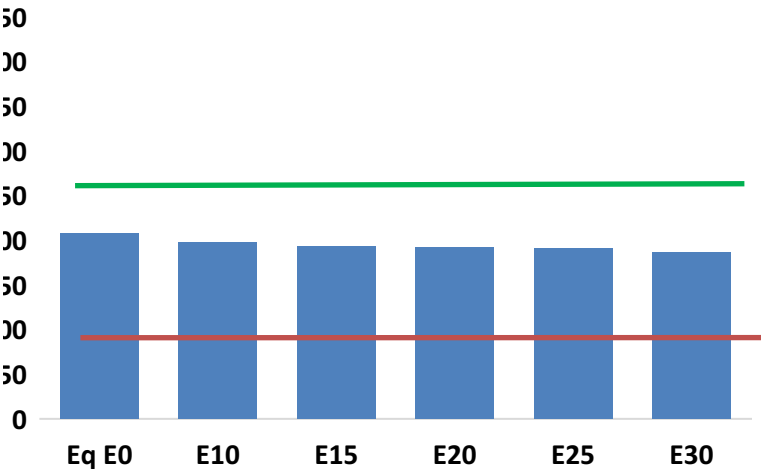
Nitrogen Oxides (NOx) g/km



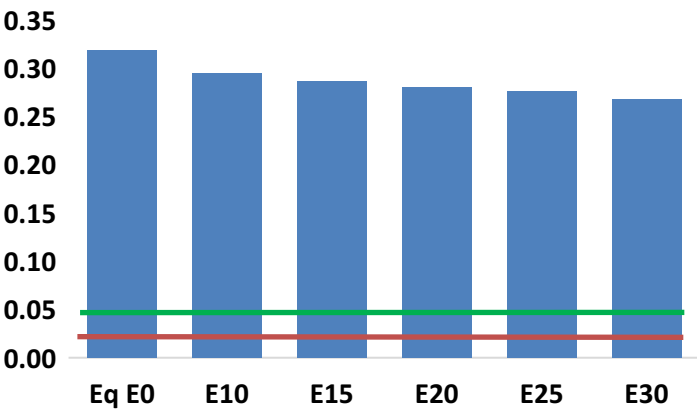
Aromatics (BTX) g/km



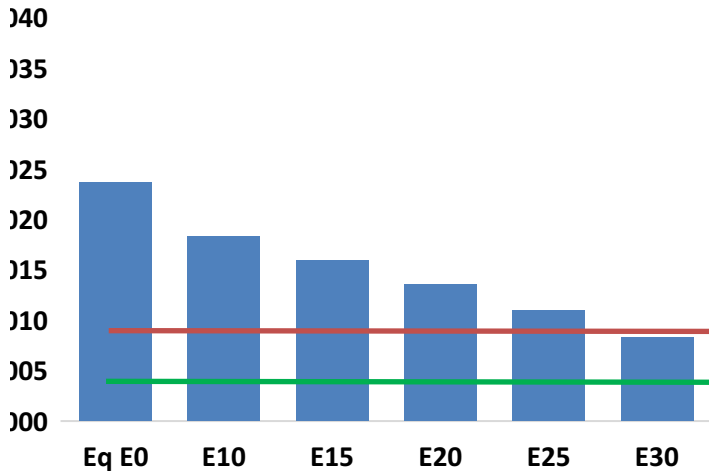
Carbon Dioxide (CO2) g/km



Volatile Organic Compounds (VOC) g/km



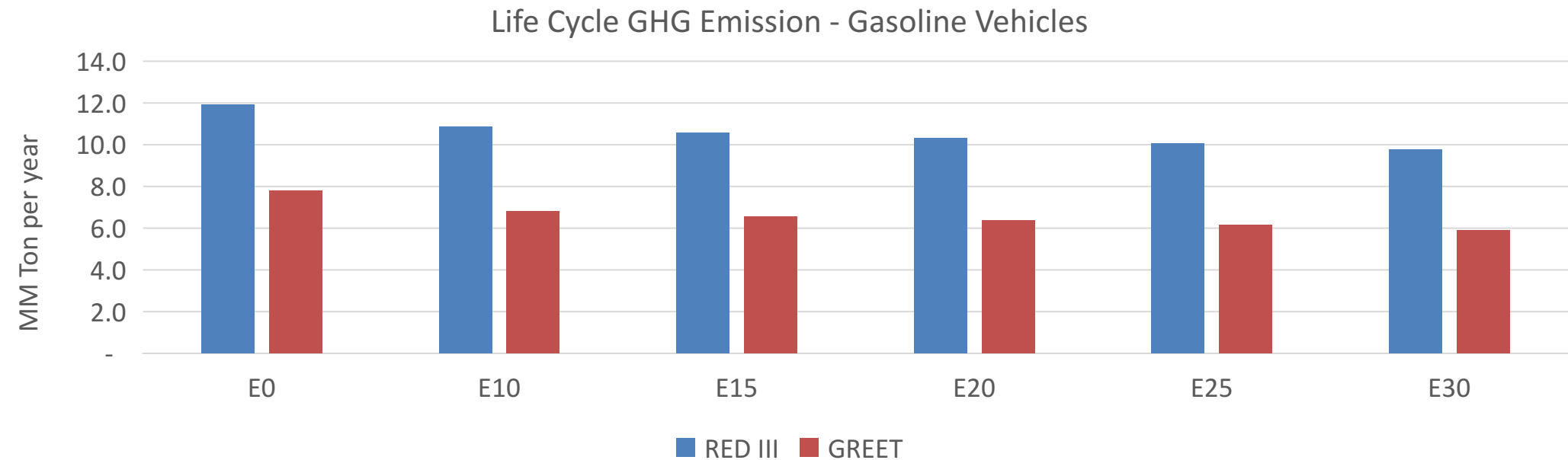
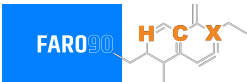
Particulate Matter (PM2.5) g/km



Portugal – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	209,074	196,659	184,244	174,785	165,594	158,626	150,367	-12%	-21%
CO2	13,460,947	13,114,808	12,787,900	12,530,796	12,403,570	12,331,260	12,103,959	-5%	-8%
VOC	20,658	19,868	19,077	18,586	18,162	17,858	17,377	-8%	-12%
NOx	17,465	13,098	12,225	11,522	10,866	10,143	9,379	-30%	-38%
SOx	160	148	136	125	116	105	94	-15%	-28%
NH3	4,599	4,553	4,508	4,529	4,580	4,616	4,646	-2%	0%
N2O	175	174	173	174	178	180	182	-1%	2%
CH4	4,900	4,900	4,900	4,974	5,082	5,175	5,263	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	198	193	187	184	182	181	178	-6%	-8%
Acetaldehyde	421	483	708	1,079	1,470	1,679	1,984	68%	249%
Formaldehyde	1,337	1,300	1,516	1,780	1,864	2,062	2,245	13%	39%
Benzene	358	344	326	321	319	305	295	-9%	-11%
PM2.5	839	745	651	566	480	390	295	-22%	-43%
PM10	690	613	536	465	395	320	243	-22%	-43%

Portugal – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	59.0
Target @ 2035 (MMT/year)	33.2
Delta	25.8

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	12.6%	15.8%	18.5%	21.0%	24.4%
RED II/III	0.0%	9.1%	11.6%	13.7%	15.7%	18.2%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.8%	4.8%	5.6%	6.4%	7.4%
RED II/III	0.0%	4.2%	5.4%	6.3%	7.3%	8.4%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90